SMART INDIA HACKATHON 2026

Vayoo

A Unified Gerontological Care Platform for
Independent, Safe and Connected Elderly Living

Problem Statement ID – SIH1778

Innovative Gerontological Care Solutions: Designing the Future of Elderly Care Products

Theme – «confirm on portal»

PS Category – Software

Team – Unmatrix

Team ID – «confirm on portal»

Working prototype — 15 database migrations, 162 tests passing

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IDEA TITLE

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Proposed Solution

- **Built and working: two paired mobile apps + one care-team console.**
- PROBLEM — an elderly parent lives alone; the family is in another city.
- SOLUTION — Vayoo connects them: emergency, health record, and consent, in one place.
- Four pillars: Emergency SOS · Health & medicines · Care marketplace · Social wellbeing.

How It Addresses the Problem

- Emergency — SOS reaches family instantly, even on a locked, force-killed phone.
- Remote care — family can order & pay for medicines without holding the account.
- Fragmentation — vitals, medicines, reports and appointments live in one record.

Innovation & Uniqueness

- Consent is the core design: senior grants access per category, revokes instantly.
- Security lives in the database, not the app — unbypassable by any client.
- **Not a concept: 15 database migrations, 162 automated security tests — already passing.**

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TECHNICAL APPROACH

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Technologies Used

- React Native (Expo) — one codebase, Android + iOS
- Next.js — the care-team console
- Supabase (PostgreSQL) — row-level security on every table
- Serverless functions — SOS dispatch, escalation, payments
- FCM + Apple Push — the alert, on both platforms
- Agora · Razorpay · MSG91 — calls, payments, SMS fallback

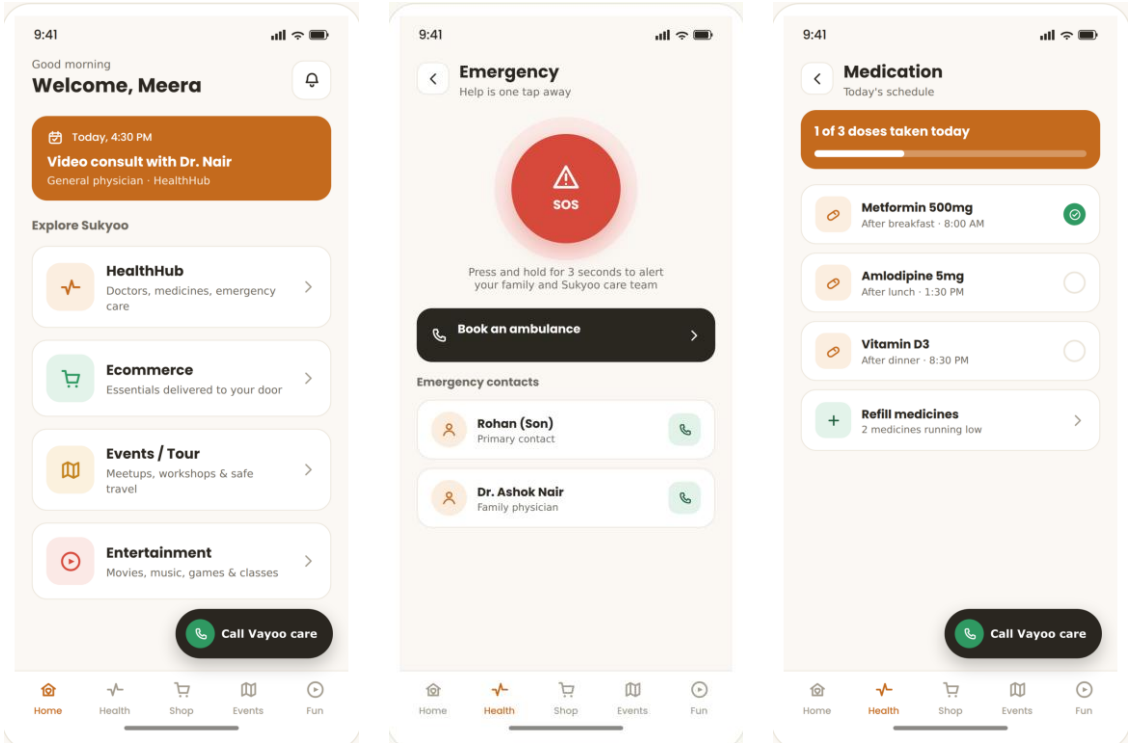
How It Works — The SOS Flow

Senior presses SOS

↓ location captured, alert raised
↓ push sent to every family device
↓ full-screen alert — wakes a locked phone
↓ no response? auto SMS / WhatsApp

An ordinary notification cannot do this — ours does.

- **Built on real security, not a demo shortcut: 162 tests attack it in CI.**



Working prototype — senior app
Home hub · Emergency SOS · Medication adherence

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FEASIBILITY AND VIABILITY

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Feasibility

- **Already built: 15 migrations, 162 tests passing, both apps compile clean.**
- Zero-cost pilot — free-tier infrastructure; real spend starts only at scale.
- Rollout through hospitals, clinics, NGOs and senior associations, city by city.
- Revenue: subscriptions + marketplace commission + partner tie-ups.

Risks → Our Strategy

- Silent alert failure → delivery is tracked; console flags it, SMS auto-fires.
- Elderly onboarding → big buttons, plain language, three text sizes.
- Data breach → database-enforced access, encrypted storage, 162 tests.
- Insider misuse → every staff health-read needs a logged reason.

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IMPACT AND BENEFITS

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Impact

- Seniors — help that reaches them, and privacy that stays theirs.
- Families — no more guessing; act on real information, from any city.
- Health system — earlier intervention, one record instead of a plastic bag.

Benefits

- Social — dignity and independence, not institutional care.
- Economic — fewer hospitalisations; local pharmacies & labs gain demand.
- Environmental — fewer repeat trips; digital records, not paper.

Policy Alignment

- Built around the DPDP Act 2023 and Telemedicine Guidelines 2020.
- Future scope: Elderline 14567, 108, and ABDM integration.

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RESEARCH AND REFERENCES

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References

- UNFPA India Ageing Report 2023 — elderly population & 2050 projection.
- Census of India 2011 — internal migration data.
- LASI (IIPS) — elderly living arrangements & health needs.
- Digital Personal Data Protection Act, 2023.
- Telemedicine Practice Guidelines, 2020 (MoHFW).
- National Policy on Older Persons — MoSJE.
- Elderline 14567 · 108 · ABDM / ABHA.

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